

Clinical Investigation

Patterns of Relapse in High-Risk Neuroblastoma Patients Treated With and Without Total Body Irradiation



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Summary

We performed an analysis of relapse patterns in patients with high-risk neuroblastoma. Patients overall showed a tendency for first relapse in anatomic sites with disease involvement at initial diagnosis. However, patients treated with systemic irradiation in the form of total body irradiation were significantly more likely to have relapse in new anatomic sites of disease rather than sites present at diagnosis.

Purpose: External beam radiation therapy to initial sites of disease may influence relapse patterns in high-risk neuroblastoma. However, the effect of systemic irradiation by use of total body irradiation (TBI) on anatomic patterns of relapse has not previously been investigated.

Methods and Materials: We retrospectively analyzed patients receiving definitive treatment of high-risk neuroblastoma with subsequent relapse in bony metastatic sites, with a date of relapse between January 1, 1997, and December 31, 2012. Anatomic sites of disease, defined by metaiodobenzylguanidine (MIBG) avidity, were compared at diagnosis and at first relapse. The Fisher exact test was performed to compare relapse in initially involved sites between patients treated with and without TBI.

Results: Seventy-four patients with a median age at diagnosis of 3.5 years (range, 0.3–15.3 years) had relapse in 227 sites of MIBG-avid metastatic disease, with a median time to relapse of 1.8 years. Of the 227 sites of first relapse, 154 sites (68%) were involved at diagnosis. When we compared relapse patterns in patients treated with and without TBI, 12 of 23 patients (52%) treated with TBI had relapse in ≥ 1 previously MIBG-avid site of disease whereas 40 of 51 patients (78%) treated without TBI had relapse in ≥ 1 previously MIBG-avid site of disease ($P = .03$).

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